Chenrui Liu

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Education

University of California, San Diego San Diego, CA
M.S. in Computer Science, GPA: 4.0/4.0 *Sep 2025 – Jun 2027 (Expected)*

Southeast University Nanjing, China
B.Eng. in Computer Science and Technology, GPA: 3.81/4.0 *Sep 2021 – Jun 2025*

- Exchange Program at Nanyang Technological University, Singapore (Jan 2025 – May 2025)
- Exchange Program at University of California, San Diego (Mar 2024 – Jun 2024)

Experience

China Telecom Hongxin Information Technology Co., Ltd. Jul 2024 – Jan 2025
Software Development Intern *Nanjing, China*

- Developed a questionnaire-driven lawsuit document generation module using Spring Boot and MyBatis-Plus.
- Implemented database-driven branching logic that dynamically selected subsequent questions based on user responses, enabling configurable decision-tree workflows.
- Designed multi-select branch processing by caching user-specific selections and sequentially resolving dependent question paths before document generation.
- Integrated Word template rendering, PDF conversion, and MinIO object storage, mapping questionnaire responses to template placeholders and enabling QR-code-based document downloads.

Projects

AI-Powered E-Commerce Support Platform Jul 2026 – Oct 2026

- Built a customer support application using FastAPI and LangGraph, enabling customers to ask about products and policies, check orders, and submit requests for human support.
- Developed a RAG pipeline using Milvus for BM25 and dense retrieval with Reciprocal Rank Fusion and reranking to generate grounded answers with citations to retrieved documents.
- Designed a 121-question evaluation set covering answerable and unsupported queries; on 106 answerable questions, achieved 0.98 Recall@5 and 0.94 MRR@10; the Qwen3.7-Flash-powered assistant correctly declined 14 of 15 unsupported questions.
- Implemented order ownership checks and customer confirmation before creating support tickets, used MySQL for business records and conversation storage, and built SSE streaming with output recovery after browser disconnection.

Paper Research Agent May 2026 – Jun 2026

- Built a local paper research chat app that helps users search arXiv, select papers, generate bibliography entries, and download PDFs through tool-calling agent workflows.
- Implemented a Python backend with OpenAI-compatible and Ollama model support, persistent SQLite chat sessions, local state management, and reliable file handling.
- Designed and developed a browser-based front end with chat history navigation, model settings panels, progress status updates, and responsive research conversation views.
- Created an evaluation harness with multi-scenario tests and packaged the application as a macOS DMG for distribution.

LLM-Based Database Schema Linking May 2026 – Jun 2026

- Built a schema-linking pipeline that predicts relevant database tables and columns from natural-language questions by prompting an instruction-tuned LLM with database schema context.
- Fine-tuned Qwen2.5-1.5B-Instruct with LoRA adapters and compared model families, learning rates, epoch counts, LoRA ranks, and target modules.
- Implemented deterministic inference and schema-aware post-processing to remove hallucinated identifiers, canonicalize casing, and reach a 0.4993 overall schema-linking score on validation.

Memory-Efficient Micro-UAV Autonomous Navigation Nov 2022 – Mar 2024

- Worked on an embedded autonomous navigation system for resource-constrained micro-UAVs using C and FreeRTOS, combining LiDAR-based 3D mapping with RRT* path planning.
- Designed a memory-efficient octree with a custom linear memory pool, allocating child nodes in contiguous groups of eight to reduce dynamic memory allocation overhead.
- Compressed each octree node into a 16-bit representation using C bit fields to encode child-unit indices, occupancy states, and leaf status, enabling efficient mapping under limited memory.
- Contributed to a granted Chinese invention patent for the system (CN116499464B).

Source code, demos, and additional projects are available on my portfolio website lllcr6.github.io.

Technical Skills

Languages: Python, Java, C/C++, SQL, JavaScript, TypeScript, HTML/CSS, MATLAB, LaTeX
Backend & APIs: Spring Boot, MyBatis-Plus, FastAPI, Flask, REST APIs, WebSocket, Server-Sent Events (SSE)
AI & Machine Learning: PyTorch, scikit-learn, LangGraph, OpenAI Agents SDK, RAG, LoRA/PEFT, Ollama
Data & Databases: MySQL, PostgreSQL, SQLite, Redis, Milvus, MinIO, NumPy, pandas, Matplotlib, Seaborn, NLTK
Tools & Infrastructure: Git, Linux, Docker, AWS, GitHub Actions, Maven, pytest, JUnit, Codex, Claude Code